Research Summary

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My research concerns integer partitions, q-series, and Rogers–Ramanujan-type identities. I combine Bailey transformations, effective coefficient estimates, and explicit partition bijections.

In joint work with Lisa Hui Sun, published in *Advances in Applied Mathematics* in 2025, I apply the Bailey lattice to derive truncated Jacobi triple product identities of Andrews–Gordon type. Specializations give new Euler and Gauss truncations and partition inequalities. The paper proves the Ballantine–Merca conjecture concerning 6-regular partitions and the case R = 3S of Merca's stronger conjecture.

A submitted analytic paper gives an explicit threshold N(R,S,k) for eventual coefficient nonnegativity when R, S, k are positive integers and $1 \le S \le R/2$. Its proof combines restricted-partition estimates with the circle method for a nonmodular infinite product. A paper accepted by the *Journal of Combinatorial Theory, Series A* proves bivariate coefficientwise nonnegativity of normalized Jacobi triple product tails for $k \ge 1$. A sign-reversing involution organizes the surviving terms by minimal excludants; an injection combines Konan's size- and length-preserving bijection with a Frobenius-arm lift. Specialization gives Merca's stronger conjecture for all $1 \le S < R$, together with inequalities for partition pairs and two-colored residue-class partitions.

Another submitted paper constructs a weight-preserving bijection between a partition-triple interpretation of the truncated pentagonal series and the Andrews–Merca partition class, using Gaussian refactorization, Durfee rectangles, and staircase operations. It also derives a conjugate well-poised Bailey pair and transformations specializing to Euler and Gauss truncations.

In joint submitted work with Ignis Duanyu Chen, Qianghui Guo and Lisa Hui Sun, we use exact counts of partitions into parts 1, 2 and 4 to extend the Ballantine–Merca inequality to regularity parameter 3 or at least 5. The proof analyzes coefficients by residue classes and gives further truncated-series inequalities.

My symmetrized B_2 Bailey preprint develops a finite kernel construction, a chain and lattice, and two families of roots. Iteration produces two-family multisums with A_1 theta products and finite theta convolutions. Selected specializations correspond to principally specialized affine A_1 characters and extremal CMPP products; a broader module realization remains a research question.

For postdoctoral work, I propose to investigate further Bailey roots and lattice directions, partition interpretations of two-family multisums, and their connections with specialized characters. I also aim to extend generalized-mex and Frobenius injections to other truncated theta series. These directions build on explicit constructions while treating new character or algebraic realizations as questions to be proved.